in GitHub Actions.

15.3.1 Understanding the production stage of the deployment pipeline

After a release candidate has gone through the commit and acceptance stages, we are confident enough to deploy it to production. The production stage can be triggered manually or automatically, depending on whether you'd like to achieve *continuous deployment*.

Continuous delivery is "a software development discipline where you build software in such a way that the software can be released to production at any time."[3] The key part is understanding that the software *can* be released to production, but it doesn't *have to*. That's a common source of confusion between continuous delivery and continuous deployment. If you also want to take the newest release candidate and deploy it to production automatically, then you would have *continuous deployment*.

The production stage consists of two main steps:

1. Update the deployment scripts (in our case, the Kubernetes manifests) with the new release version.
2. Deploy the application to the production environment.

NOTE An optional third step would be to run some final automated tests to verify that the deployment was successful. Perhaps you could reuse the same system tests that you will have included in the acceptance stage to verify the deployment in a staging environment.

The next section will show you how to implement the first step of the production stage using GitHub Actions, and we'll discuss some implementation strategies for the second step. We'll aim to automate the